

CVP1N: IR1101 Wired SD-Routing with LTE as Last Resort and DIA

Overview

This single-router SD-Routing configuration is for the Cisco Industrial Router IR1101. It is designed specifically for this platform with all its unique interface names and modules to aid in the rapid deployments at small to medium sized locations where flexibility, simplicity and standardization are key. This catalog represents a Cisco Validated Profile (CVP) suited for deployments at locations like retail, Oil & Gas Pipelines, Roadways & Intersections.

This network configuration is wired transport with single LTE connectivity as last resort. This configuration offers network segmentation via 1 secure VPN and NAT functionality to all connected devices for internet access over VPN 0 using Direct Internet Access (DIA).

Configuration Catalog Details

Ethernet LAN connectivity is mapped into one VPN for security and granular control:

- Connected Devices (VPN 10): Secure network or DIA for corporate devices.
- All edge devices can reach the internet through NAT using VPN 0. If the user wants to route some traffic from edge to enterprise network, then a hub needs to be deployed using a configuration group not provided and also using a routed SDR catalog with the appropriate IPsec config, refer to catalog CVP1R_SDR for this functionality.

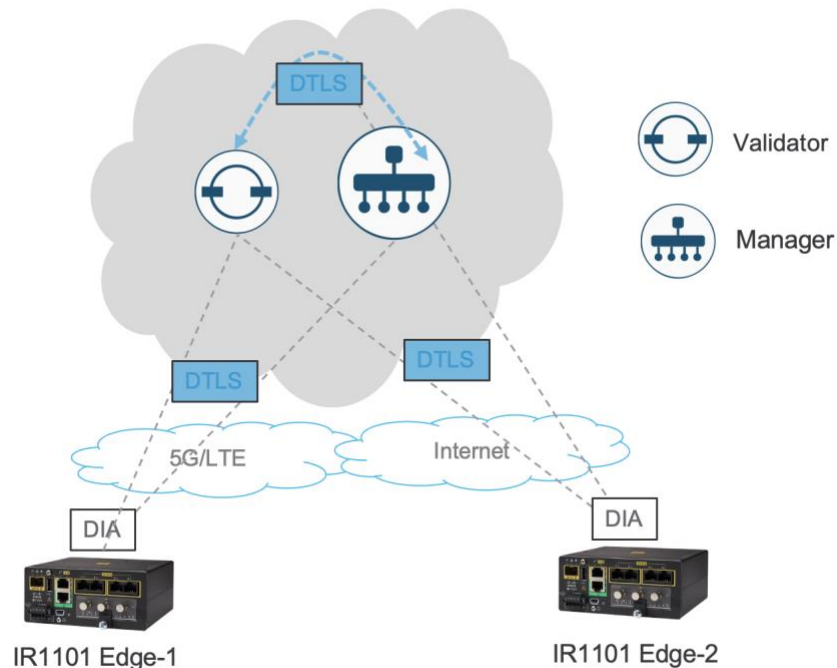


Fig 1: Configuration Catalog Design

Further details on functionality provided by this catalog entry which can be changed for the device at deployment time:

- The Service VPN (LAN) has one VRF configured with one SVI and 1 Loopback interface. All default route traffic from VPN 10 will traverse the VPN 0 through NAT to the public internet.
- A DHCP Server is configured for VPN 10 allowing connected clients to receive IP addresses from service VPN 10.
- A primary and secondary NTP servers with corresponding VPN for access can be configured with primary as preferred source. Also, NTP on wired can be disabled if needed.
- Default DNS servers (Cisco DNS) are provided in both VPN 0 and VPN 10. User can change those in VPN 10 as needed.
- Point router logs to a server of their choice and provide the correct VPN and source interface for such traffic. The field is in the system section called "Logging Server IP Address" and is currently set to "0.0.0.0", please change it at install time to a valid logging server IP address or set it to be the same as the "System IP" from the system section if one does not exist.
- Enable GPS coordinates for the device to display in UI/MAP if the device has LTE modem with GPS. Both GPS and NMEA must be enabled and the mode set to "standalone" to acquire GPS signal. Modem GPS coordinates will override static coordinates in UI/MAP. If NMEA data streaming is also required, source/destination IP and port should be provided to forward the GPS stream.

Cisco provides the configurations in this catalog as is for your convenience. These configurations have been built using industry best practices, observed across multiple deployments, which may be beneficial to you. Cisco is not responsible for any technical issues, bugs, or other issues that may arise from your use of these configurations and any resulting indirect, incidental, reliance, consequential, special or exemplary damages or loss of actual or anticipated revenue, profit, business, savings, data goodwill or use, business interruption, damaged data, wasted expenditure or delay in delivery (in all cases, whether direct or indirect).